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Semiconductor device including structure connecting frontside and backside metal and method of manufacturing the same

Abstract

The present disclosure provides a semiconductor device and a method of manufacturing a semiconductor device. The semiconductor device comprises a substrate, an isolation layer, a first electronic device, a first interconnection structure, a first conductive structure, and a second conductive structure. The substrate has a first surface and a second surface opposite the first surface. The isolation layer contacts the second surface of the substrate and has a first surface facing away from the substrate. The first electronic device is embedded in the substrate. The first interconnection structure extends from the first surface of the substrate to the first surface of the isolation layer. The first conductive structure is disposed on the first surface of the substrate. The second conductive structure contacts the first surface of the isolation layer. The first conductive structure and the second conductive structure are electrically connected by the first interconnection structure.

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Background/Summary

BACKGROUND

(1) The present invention relates generally to semiconductor devices, and more particularly to semiconductor devices connecting frontside and backside metals for signal routing and power delivery network and methods of manufacturing the same.

(2) Scaled technologies put considerable strain on back end of line (BEOL) design, leading to multiple challenges like pin access and routing congestion. In addition, with dimension scaling, IC

performance becomes limited by BEOL resistivity. Conventionally only frontside (F/S) BEOL is used for signal and power delivery network (PDN) routing on the chip. The tight pitch required for the lower metal layers of the F/S BEOL leads to high resistivity and capacitance in the metal routing due to shallow metal and the proximity of such metal layers.

(3) Innovative technologies are therefore needed to tackle these bottlenecks to assist further device and IC scaling. Backside (B/S) metal layers are separately manufactured under the substrate, which allows B/S metal customization in terms of metal pitch, thickness, and critical dimension independent of design rules in the F/S BEOL, which enables flexible optimization of metal RC. However, a key challenge is connection of F/S and B/S metal with a low-resistance path.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the embodiments of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various structures are not drawn to scale. In fact, the dimensions of the various structures may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1A illustrates an exemplary top view of a semiconductor structure, in accordance with some embodiments of the present disclosure.

(3) FIG. 1B illustrates an exemplary cross-section of a semiconductor structure, in accordance with some embodiments of the present disclosure.

(4) FIG. 2A illustrates an exemplary top view of a semiconductor structure, in accordance with some embodiments of the present disclosure.

(5) FIG. 2B illustrates an exemplary cross-section of a semiconductor structure, in accordance with some embodiments of the present disclosure.

(6) FIG. 3A illustrates an exemplary top view of a semiconductor structure, in accordance with some embodiments of the present disclosure.

(7) FIG. 3B illustrates an exemplary cross-section of a semiconductor structure, in accordance with some embodiments of the present disclosure.

(8) FIG. 4 illustrates an exemplary cross-section of a semiconductor structure, in accordance with some embodiments of the present disclosure.

(9) FIG. 5A, FIG. 5B, FIG. 5C, FIG. 5D, and FIG. 5E illustrate various stages of manufacturing a semiconductor device structure, in accordance with some embodiments of the present disclosure.

(10) FIG. 6A, FIG. 6B, FIG. 6C, and FIG. 6D illustrate various stages of manufacturing a semiconductor device structure, in accordance with some embodiments of the present disclosure.

(11) FIG. 7 is a flowchart illustrating a method for manufacturing a semiconductor device, in accordance with various aspects of the present disclosure.

DETAILED DESCRIPTION

(12) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of elements and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments

and/or configurations discussed.

(13) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “over,” “upper,” “on” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(14) As used herein, although terms such as “first,” “second” and “third” describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may only be used to distinguish one element, component, region, layer or section from another. Terms such as “first,” “second” and “third” when used herein do not imply a sequence or order unless clearly indicated by the context.

(15) Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the disclosure are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in the respective testing measurements.

Also, as used herein, the terms “substantially,” “approximately” and “about” generally mean within a value or range that can be contemplated by people having ordinary skill in the art. Alternatively, the terms “substantially,” “approximately” and “about” mean within an acceptable standard error of the mean when considered by one of ordinary skill in the art. People having ordinary skill in the art can understand that the acceptable standard error may vary according to different technologies.

Other than in the operating/working examples, or unless otherwise expressly specified, all of the numerical ranges, amounts, values and percentages such as those for quantities of materials, durations of times, temperatures, operating conditions, ratios of amounts, and the likes thereof disclosed herein should be understood as modified in all instances by the terms “substantially,” “approximately” or “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the present disclosure and attached claims are approximations that can vary as desired. At the very least, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Ranges can be expressed herein as from one endpoint to another endpoint or between two endpoints. All ranges disclosed herein are inclusive of the endpoints, unless specified otherwise.

(16) FIG. 1A illustrates an exemplary top view of a semiconductor structure, in accordance with some embodiments of the present disclosure. From a top view perspective of a semiconductor structure, a tap cell **100** is a region wherein a feedthrough via is formed. Conductive structures **101** and **101'** are used for transmitting power signals for the semiconductor structure. The conductive structure **101** comprises a portion **101s** and another portion **101j** integrally formed. The portion **101s** can be referred to as a straight portion, while the portion **101j** can be referred to as a jog portion. An interconnection structure **102** may be a feedthrough via (FTV). The conductive structure **101** is electrically connected to the interconnection structure **102** through the portion **101j**. In some embodiments, the interconnection structure **102** is disposed between the conductive structures **101** and **101'**. In some embodiments, the portion **101j** can cover the interconnection structure **102** from the top view perspective. In other embodiments, the portion **101j** may expose at least a portion of the interconnection structure **102** from the top view perspective.

(17) As shown in FIG. 1A, the tap cell **100** includes a part of the portion **101s**, the portion **101j**, a part of the conductive structure **101'** and the interconnection structure **102**. The tap cell has a pitch **103**. The pitch **103** is in a range of 0.03 to 300 μm . In the present disclosure, a tap cell can be a zone in which no active semiconductor device is disposed. The interconnection structure **102** arranged in the tap cell **100** can thus be kept a sufficient distance from neighboring active semiconductor devices.

(18) A conductive structure **104** comprises fingers **104f1-104f5**. The fingers **104f1-104f5** can collaboratively form a fence pattern. The fingers **104f1-104f5** are parallel to the conductive structures **101** and **101'**. The conductive structure **104** is disposed between the conductive structures **101** and **101'**. The conductive structure **104** can be configured for transmitting signals. Each of the fingers **104f1-104f5** can be referred to as a signal track. The tap cell **100** has a cell height **105**. The cell height **105** is in a range of 40 to 300 nm.

(19) FIG. 1B illustrates an exemplary cross-section of a semiconductor structure **10**, in accordance with some embodiments of the present disclosure. The cross-section shown in FIG. 1B can correspond to the cross-section obtained along the dashed line A-A' of FIG. 1A. A substrate **106** is provided. The substrate **106** has a surface **106a** and a surface **106b** opposite the surface **106a**. An isolation layer **107** is formed under the surface **106b**. The isolation layer **107** contacts the surface **106b** of the substrate **106**. The isolation layer **107** has a surface **107a** facing away from the substrate **106**. The semiconductor structure **10** includes electronic devices **108** and **109** embedded in the substrate **106**. The electronic devices **108** and **109** may be of different conductive types. An interconnection structure **102** extends from the surface **106a** of the substrate **106** to the surface **107a** of the isolation layer **107**. The interconnection structure **102** penetrates the substrate **106** and the isolation layer **107**. One terminal of the interconnection structure **102** can be exposed by the surface **106a** of the substrate **106**, and the other terminal of the interconnection structure **102** can be exposed by the surface **107a** of the isolation layer **107**.

(20) An isolation layer **111** is disposed on the surface **106a** of the substrate **106**. An isolation layer **112** is disposed on the isolation layer **111**. A conductive structure **101** is disposed on the surface **106a** of the substrate **106**. The conductive structure **101** is sandwiched between the isolation layers **111** and **114**. The conductive structure **101** is embedded in the isolation layer **112**. The conductive structure **101** has a surface **101a** facing away from the substrate **106**. A conductive structure **113** is disposed on the surface **106a** of the substrate **106**. The conductive structure **113** is sandwiched between the isolation layers **111** and **114**. The conductive structure **113** is embedded in the isolation layer **112**. An isolation layer **114** is disposed on the surface **101a** of the conductive structure **101**. The isolation layer **114** is disposed on the isolation layer **112**. An isolation layer **115** is disposed on the isolation layer **114**. A conductive structure **116** is embedded in the material **115**. An interconnection structure **110** electrically connects the conductive structure **101** and the electronic device **108**.

(21) An isolation layer **117** is disposed under the surface **107a** of the isolation layer **107**. An isolation layer **118** is disposed under the isolation layer **117**. A conductive structure **119** is disposed under the surface **107a** of the isolation layer **107**. The conductive structure **119** is embedded in the isolation layer **118**. The conductive structure **119** contacts the surface **107a** of the isolation layer **107**. A conductive structure **120** is disposed under the surface **107a** of the isolation layer **107**. The conductive structure **120** is embedded in the isolation layer **118**. The conductive structure **120** contacts the surface **107a** of the isolation layer **107**. The conductive structure **101** and the conductive structure **119** are electrically connected by the interconnection structure **102**.

(22) As shown in FIG. 1B, the electronic device **108** and the electronic device **109** are disposed on different sides of the interconnection structure **102**. The conductive structure **101** and the conductive structure **119** are configured to receive a first reference voltage (e.g., VDD). The electronic device **108** can receive the first reference voltage through, for example, the conductive structure **119**, the interconnection structure **102**, the conductive structure **101**, and the interconnection structure **110**. Referring to FIG. 1B, the PDN for delivering power to the electronic device **108** involves conductive structures on both sides of the substrate **106** (i.e., the conductive structures **101** and **119**).

(23) The conductive structure **113** and the conductive structure **120** are configured to receive a second reference voltage (e.g., VSS). The conductive structure **113** is spaced apart from the conductive structure **101**. The conductive structure **120** is spaced apart from the conductive

structure **119**. The conductive structure **116** is separated from the conductive structure **101** and the conductive structure **113**.

(24) The interconnection structure **102** comprises a body **102b** and a barrier layer **102a**. The barrier layer **102a** is disposed on sidewalls, a top surface, and a bottom surface of the body **102b**. The barrier layer **102a** comprises one or more of Ta, TaN, Co, Ru, Al, Ti, TiN, Mn, Nb or air. The body **102b** comprises one or more of Cu, Ru, W, Ti, Al, Co, Mo, Ir, Rh.

(25) FIG. 2A illustrates an exemplary top view of a semiconductor structure, in accordance with some embodiments of the present disclosure. From a top view perspective of a semiconductor structure, a tape cell **200** is a region wherein a feedthrough via is formed. Conductive structures **201** and **201'** are used for transmitting power signals for the semiconductor structure. An interconnection structure **202**, which may be a FTV, is disposed within the tap cell **200**. In some embodiments, the interconnection structure **202** is disposed between the conductive structures **201** and **201'**. The interconnection structure **202** can be sandwiched between the conductive structures **201** and **201'** from the top view perspective.

(26) As shown in FIG. 2A, the tap cell has a pitch **203**. The pitch **203** is in a range of 0.03 to 3000 μm . A conductive structure **204** comprises fingers **204f1-204f5**. The fingers **204f1-204f5** can collaboratively form a fence pattern. The conductive structure **204** is disposed between the conductive structures **201** and **201'**. The fingers **204f1-204f5** partially covers the interconnection structure **202** from the top view perspective. The fingers **204f1-204f5** expose at least a portion of the interconnection structure **202** from the top view perspective. The fingers **204f1-204f5** are parallel to the conductive structures **201** and **201'**. The conductive structure **205** is disposed between the conductive structures **201** and **201'**. The conductive structure **205** comprises a fence pattern. The conductive structures **204** and **205** are used for transmitting signals. Each of the fingers of the conductive structures **204** and **205** can be referred to as a signal track. The tap cell **200** has a cell height **206**. The cell height **206** is in a range of 40 to 300 nm.

(27) From the top view perspective, a conductive structure **207** partially covers the conductive structures **201** and **201'**, the interconnection structure **202**, and the conductive structure **204**. The conductive structure **207** is orthogonal to the conductive structures **201** and **201'** and the conductive structure **204**. The conductive structure **207** has vias **207v1-207v4**. The via **207v1** is disposed at the intersection of the conductive structure **207** and the conductive structure **201**. The vias **207v2-207v4** are disposed at the intersection of the conductive structure **207** and the conductive structure **204**. As shown in FIG. 2A, the tap cell **200** includes a part of the conductive structures **201**, a part of the conductive structure **201'**, the interconnection structure **202**, the conductive structure **204** and a portion of the conductive structure **207**.

(28) FIG. 2B illustrates an exemplary cross-section of a semiconductor structure **20**, in accordance with some embodiments of the present disclosure. The cross-section shown in FIG. 2B can correspond to the cross-section obtained along the dashed line B-B' of FIG. 2A. A substrate **208** is provided. The substrate **208** has a surface **208a** and a surface **208b** opposite the surface **208a**. An isolation layer **209** is formed under the surface **208b**. The isolation layer **209** contacts the surface **208b** of the substrate **208**. The isolation layer **209** has a surface **209a** facing away from the substrate **208**. The semiconductor structure **20** includes electronic devices **210** and **211** embedded in the substrate **208**. The electronic devices **210** and **211** may be of different conductive types. An interconnection structure **202** extends from the surface **208a** of the substrate **208** to the surface **209a** of the isolation layer **209**. The interconnection structure **202** penetrates the substrate **208** and the isolation layer **209**. One terminal of the interconnection structure **202** can be exposed by the surface **208a** of the substrate **208**, and the other terminal of the interconnection structure **202** can be exposed by the surface **209a** of the isolation layer **209**.

(29) An isolation layer **213** is disposed on the surface **208a** of the substrate **208**. An isolation layer **214** is disposed on the isolation layer **213**. A conductive structure **201** is disposed on the surface **208a** of the substrate **208**. The conductive structure **201** is sandwiched between the isolation layers

213 and **215**. The conductive structure **201** is embedded in the isolation layer **214**. The conductive structure **201** has a surface **201a** facing away from the substrate **208**. A conductive structure **215** is disposed on the surface **208a** of the substrate **208**. The conductive structure **215** is sandwiched between the isolation layers **213** and **215**. The conductive structure **215** is embedded in the isolation layer **214**. A conductive structure **204** is disposed on the surface **208a** of the substrate **208**. The conductive structure **204** comprises fingers **204f1-204f5**. The fingers **204f1-204f5** can collaboratively form a fence pattern. The conductive structure **204** covers a first portion of the interconnection structure **202** and exposes a second portion of the interconnection structure **202**.

(30) An isolation layer **216** is disposed on the surface **201a** of the conductive structure **201**. The isolation layer **216** is disposed on the isolation layer **214**. An isolation layer **217** is disposed on the isolation layer **216**. A conductive structure **207** is embedded in the isolation layer **217**. The conductive structure **207** comprises vias **207v1-207v4**. The vias **207v1-207v4** are disposed on the surface **201a** of the conductive structure **201**. The via **207v1** contacts the conductive structure **201**. The via **207v2** contacts the fence pattern **204f2** of the conductive structure **204**. The via **207v3** contacts the fence pattern **204f3** of the conductive structure **204**. The via **207v4** contacts the fence pattern **204f4** of the conductive structure **204**. An interconnection structure **212** electrically connects the conductive structure **201** and the electronic device **210**.

(31) An isolation layer **218** is disposed under the surface **209a** of the isolation layer **209**. An isolation layer **219** is disposed under the isolation layer **218**. A conductive structure **220** is disposed under the surface **209a** of the isolation layer **209**. The conductive structure **220** is embedded in the isolation layer **219**. The conductive structure **220** contacts the surface **209a** of the isolation layer **209**. A conductive structure **221** is disposed under the surface **209a** of the isolation layer **209**. The conductive structure **221** is embedded in the isolation layer **219**. The conductive structure **221** contacts the surface **209a** of the isolation layer **209**. The conductive structure **207** and the conductive structure **220** are electrically connected by the interconnection structure **202** and the conductive structure **204**.

(32) As shown in FIG. 2B, the electronic device **210** and the electronic device **211** are disposed on different sides of the interconnection structure **202**. The conductive structure **201** and the conductive structure **220** are configured to receive a first reference voltage (e.g., VDD). The electronic device **210** can receive the first reference voltage through, for example, the conductive structure **220**, the interconnection structure **202**, the conductive structure **204**, the conductive structure **207**, the conductive structure **201**, and the interconnection structure **212**. Referring to FIG. 2B, the PDN for delivering power to the electronic device **210** involves conductive structures on both sides of the substrate **208** (i.e., the conductive structures **201**, **204**, **207**, and **220**).

(33) The conductive structure **215** and the conductive structure **221** are configured to receive a second reference voltage (e.g., VSS). The conductive structure **215** is spaced apart from the conductive structure **201**. The conductive structure **221** is spaced apart from the conductive structure **220**. The conductive structure **204** is spaced apart from the conductive structures **201** and **215**.

(34) The interconnection structure **202** comprises a body **202b** and a barrier layer **202a**. The barrier layer **202a** is disposed on sidewalls and a bottom surface of the body **202b**. The barrier layer **202a** comprises one or more of Ta, TaN, Co, Ru, Al, Ti, TiN, Mn, Nb or air. The body **202b** comprises one or more of Cu, Ru, W, Ti, Al, Co, Mo, Ir, Rh.

(35) FIG. 3A illustrates an exemplary top view of a semiconductor structure, in accordance with some embodiments of the present disclosure. From a top view perspective of a semiconductor structure, a tape cell **300** is a region wherein a feedthrough via is formed. Conductive structures **301** and **301'** are used for transmitting power signals for the semiconductor structure. In some embodiments, a conductive structure **302** is disposed between the conductive structures **301** and **301'**. The conductive structure **302** is spaced apart from the conductive structures **301** and **301'** from the top view. In some embodiments, an interconnection structure **303**, which may be a FTV, is

disposed within the tap cell **300**. In some embodiments, the interconnection structure **303** is disposed between the conductive structures **301** and **301'**. The interconnection structure **303** can be sandwiched between the conductive structures **301** and **301'** from the top view perspective.

(36) As shown in FIG. 3A, the tap cell has a pitch **304**. The pitch **304** is in a range of 0.03 to 3000 μm . A conductive structure **305** comprises fingers **305f1-305f5**. The fingers **305f1-305f5** can collaboratively form a fence pattern. The conductive structure **302** is disposed between the conductive structures **301** and **301'**. The fingers **305f1-305f5** are parallel to the conductive structures **301** and **301'**. The conductive structure **303** is disposed between the conductive structures **301** and **301'**. The conductive structure **305** can be configured for transmitting signals. Each of the fingers **305f1-305f5** can be referred to as a signal track. The tap cell **300** has a cell height **306**. The cell height **306** is in a range of 40 to 300 nm.

(37) From the top view perspective, a conductive structure **307** partially covers the conductive structures **301** and **301'**, the conductive structure **302**, and the interconnection structure **303**. The conductive structure **307** is orthogonal to the conductive structures **301** and **301'**. The conductive structure **307** has vias **307v1-307v3**. The vias **307v1-307v3** are disposed at the intersection of the conductive structure **307** and the conductive structure **302**. As shown in FIG. 2A, the tap cell **300** includes a part of the conductive structures **301**, a part of the conductive structure **301'**, the conductive structure **302**, the interconnection structure **303** and a portion of the conductive structure **307**.

(38) FIG. 3B illustrates an exemplary cross-section of a semiconductor structure **30**, in accordance with some embodiments of the present disclosure. The cross-section shown in FIG. 3B can correspond to the cross-section obtained along the dashed line C-C' of FIG. 3A. A substrate **308** is provided. The substrate **308** has a surface **308a** and a surface **308b** opposite the surface **308a**. An isolation layer **309** is formed under the surface **308b**. The isolation layer **309** contacts the surface **308b** of the substrate **308**. The isolation layer **309** has a surface **309a** facing away from the substrate **308**. The semiconductor structure **30** includes electronic devices **310** and **311** embedded in the substrate **308**. The electronic devices **310** and **311** may be of different conductive types.

(39) An interconnection structure **303** extends from the surface **308a** of the substrate **308** to the surface **309a** of the isolation layer **309**. The interconnection structure **303** has a first end exposed by the surface **308a** of the substrate **308** and a second end exposed by the surface **309a** of the isolation layer **309**. The interconnection structure **303** penetrates the substrate **308** and the isolation layer **309**. The interconnection structure **303** can be configured to transmit signals. The electronic device **310** may receive power from the conductive structure **301**. The electronic device **310** may receive signal from the interconnection structure **303**.

(40) An isolation layer **313** is disposed on the surface **308a** of the substrate **308**. An isolation layer **314** is disposed on the isolation layer **313**. A conductive structure **301** is disposed on the surface **308a** of the substrate **308**. The conductive structure **301** is embedded in the isolation layer **314**. The conductive structure **301** has a surface **301a** facing away from the substrate **308**. A conductive structure **315** is disposed on the surface **308a** of the substrate **308**. The conductive structure **315** is embedded in the isolation layer **314**. A conductive structure **302** is disposed on the surface **308a** of the substrate **308**. The conductive structure **302** is embedded in the isolation layer **314**.

(41) An isolation layer **316** is disposed on the surface **301a** of the conductive structure **301**. The isolation layer **316** is disposed on the isolation layer **314**. An isolation layer **317** is disposed on the isolation layer **316**. A conductive structure **307** is disposed on the surface **308a** of the substrate **308**. The conductive structure **307** comprises vias **307v1-307v3**. The vias **307v1-307v3** are disposed on the surface **301a** of the conductive structure **301**. The via **307v1** contacts the conductive structure **302**. The via **307v2** contacts the conductive structure **302**. The via **307v3** contacts the conductive structure **302**. An interconnection structure **312** electrically connects the conductive structure **301** and the electronic device **310**.

(42) An isolation layer **318** is disposed under the surface **309a** of the isolation layer **309**. An

isolation layer **319** is disposed under the isolation layer **318**. A conductive structure **320** is disposed under the surface **309a** of the isolation layer **309**. The conductive structure **320** is embedded in the isolation layer **319**. The conductive structure **320** contacts the surface **309a** of the isolation layer **309**. A conductive structure **321** is disposed under the surface **309a** of the isolation layer **309**. The conductive structure **321** is embedded in the isolation layer **319**. The conductive structure **321** contacts the surface **309a** of the isolation layer **309**. A conductive structure **322** is disposed under the surface **309a** of the isolation layer **309**. The conductive structure **322** is embedded in the isolation layer **319**. The conductive structure **322** contacts the surface **309a** of the isolation layer **309**. The conductive structure **322** contacts the interconnection structure **303**. The conductive structure **302** and the conductive structure **322** are electrically connected by the interconnection structure **303**.

(43) As shown in FIG. 3B, the electronic device **310** and the electronic device **311** are disposed on different sides of the interconnection structure **303**. The conductive structure **301** and the conductive structure **320** are configured to receive a first reference voltage (e.g., VDD). The conductive structure **315** and the conductive structure **321** are configured to receive a second reference voltage (e.g., VSS). The conductive structure **315** is spaced apart from the conductive structure **301**. The conductive structure **321** is spaced apart from the conductive structure **320**. The conductive structure **302** is spaced apart from the conductive structures **301** and **315**. The conductive structure **322** is spaced apart from the conductive structure **320** and **321**.

(44) The conductive structures **302** and **322** are configured to receive a signal. In some embodiments, the interconnection structure **303** is used for transmitting signals, such as, for example, control signals for the electronic device **310** and/or the electronic device **311**. In some embodiments, the transmission in the interconnection structure **303** can be periodic or aperiodic signals. The electronic device **310** may receive power from the conductive structure **301**. The electronic device **310** may receive signals from the interconnection structure **303**.

(45) The interconnection structure **303** comprises a body **303b** and a barrier layer **303a**. The barrier layer **303a** is disposed on sidewalls and a bottom surface of the body **303b**. The barrier layer **303a** of the interconnection structure **303** comprises one or more of Ta, TaN, Co, Ru, Al, Ti, TiN, Mn, Nb or air. The body **303b** of the interconnection structure **303** comprises one or more of Cu, Ru, W, Ti, Al, Co, Mo, Ir, Rh.

(46) The conductive structure **302** comprises a body **302b** and a barrier layer **302a**. The barrier layer **302a** is disposed on sidewalls and a bottom surface of the body **302b**. The barrier layer **302a** of the conductive structure **302** comprises one or more of Ta, TaN, Co, Ru, Al, Ti, TiN, Mn, Nb or air. The body **302b** of the conductive structure **302** comprises one or more of Cu, Ru, W, Ti, Al, Co, Mo, Ir, Rh.

(47) As shown in FIG. 3B, a distance **H1** from the surface **308a** of the substrate **308** to the surface **309a** of the isolation layer **309** is in a range of about 100 to about 1000 nm. A distance **W1** between one sidewall on one side of the interconnection structure **303** and the other sidewall on the other side of the interconnection structure **303** is in a range of about 20 to about 200 nm.

(48) FIG. 4 illustrates an exemplary cross-section of a semiconductor structure **40**, in accordance with some embodiments of the present disclosure. The cross-section shown in FIG. 4 can correspond to the cross-section obtained along the dashed line C-C' of FIG. 3A. A substrate **408** is provided. The substrate **408** has a surface **408a** and a surface **408b** opposite the surface **408a**. An isolation layer **409** is formed under the surface **408b**. The isolation layer **409** contacts the surface **408b** of the substrate **408**. The isolation layer **409** has a surface **409a** facing away from the substrate **408**. The semiconductor structure **40** includes electronic devices **410** and **411** embedded in the substrate **408**. The electronic devices **410** and **411** may be of different conductive types. An interconnection structure **403** extends from the surface **408a** of the substrate **408** to the surface **409a** of the isolation layer **409**. One terminal of the interconnection structure **403** can be exposed by the surface **408a** of the substrate **408**, and the other terminal of the interconnection structure **403**

can be exposed by the surface **409a** of the isolation layer **409**. The interconnection structure **403** can be configured to transmit signals. The electronic device **410** may receive power from the conductive structure **401**. The electronic device **410** may receive signals from the interconnection structure **403**.

(49) An isolation layer **413** is disposed on the surface **408a** of the substrate **408**. An isolation layer **414** is disposed on the isolation layer **413**. A conductive structure **401** is disposed on the surface **408a** of the substrate **408**. The conductive structure **401** is embedded in the isolation layer **414**. The conductive structure **401** has a surface **401a** facing away from the substrate **408**. A conductive structure **415** is disposed on the surface **408a** of the substrate **408**. The conductive structure **415** is embedded in the isolation layer **414**. A conductive structure **402** is disposed on the surface **408a** of the substrate **408**. The conductive structure **402** is embedded in the isolation layer **414**.

(50) An isolation layer **416** is disposed on the surface **401a** of the conductive structure **401**. The isolation layer **416** is disposed on the isolation layer **414**. An isolation layer **417** is disposed on the isolation layer **416**. A conductive structure **407** is disposed on the surface **408a** of the substrate **408**. The conductive structure **407** comprises vias **407v1-407v3**. The vias **407v1-407v3** are disposed on the surface **401a** of the conductive structure **401**. The via **407v1** contacts the conductive structure **402**. The via **407v2** contacts the conductive structure **402**. The via **407v3** contacts the conductive structure **402**. An interconnection structure **412** electrically connects the conductive structure **401** and the electronic device **410**. The vias **407v1-407v3** of the conductive structure **407** covers a first portion of the conductive structure **402** and exposes a second portion of the conductive structure **402**.

(51) An isolation layer **418** is disposed under the surface **409a** of the isolation layer **409**. An isolation layer **419** is disposed under the isolation layer **418**. A conductive structure **420** is disposed under the surface **409a** of the isolation layer **409**. The conductive structure **420** is embedded in the isolation layer **419**. The conductive structure **420** contacts the surface **409a** of the isolation layer **409**. A conductive structure **421** is disposed under the surface **409a** of the isolation layer **409**. The conductive structure **421** is embedded in the isolation layer **419**. The conductive structure **421** contacts the surface **409a** of the isolation layer **409**. A conductive structure **422** is disposed under the surface **409a** of the isolation layer **409**. The conductive structure **422** is embedded in the isolation layer **419**. The conductive structure **422** contacts the surface **409a** of the isolation layer **409**. The conductive structure **422** contacts the interconnection structure **403**. The conductive structure **402** and the conductive structure **422** are electrically connected by the interconnection structure **403**.

(52) An isolation layer **423** is disposed under a surface **419a** of the isolation layer **419**. An isolation layer **424** is disposed under the isolation layer **423**. A conductive structure **425** is disposed under the surface **419a** of the isolation layer **419**. The conductive structure **425** is embedded in the isolation layer **424**. The conductive structure **425** contacts the surface **419a** of the isolation layer **419**. The conductive structure **425** contacts the conductive structure **422**. The conductive structure **425** comprises a via **425v1**. The via **425v1** is disposed under the surface **419a** of the isolation layer **419**. The via **425v1** contacts the conductive structure **422**. The conductive structure **407** and the conductive structure **425** are electrically connected by the conductive structure **402**, the interconnection structure **403** and conductive structure **422**.

(53) As shown in FIG. 4, the electronic device **410** and the electronic device **411** are disposed on different sides of the interconnection structure **403**. The conductive structure **401** and the conductive structure **420** are configured to receive a first reference voltage (e.g., VDD). The conductive structure **415** and the conductive structure **421** are configured to receive a second reference voltage (e.g., VSS). The conductive structure **415** is spaced apart from the conductive structure **401**. The conductive structure **421** is spaced apart from the conductive structure **420**. The conductive structure **402** is spaced apart from the conductive structures **401** and **415**. The conductive structure **422** is spaced apart from the conductive structure **420** and **421**. The

conductive structure **402**, **422** and **425** are configured to receive a signal. In some embodiments, the interconnection structure **403** is used for transmitting signals. The electronic device **410** receive power from the conductive structure **401**. The electronic device **410** may receive signals from the interconnection structure **403**.

(54) The interconnection structure **403** comprises a body **403b** and a barrier layer **403a**. The barrier layer **403a** is disposed on sidewalls and a bottom surface of the body **403b**. The barrier layer **403a** of the interconnection structure **403** comprises one or more of Ta, TaN, Co, Ru, Al, Ti, TiN, Mn, Nb or air. The body **403b** of the interconnection structure **403** comprises one or more of Cu, Ru, W, Ti, Al, Co, Mo, Ir, Rh.

(55) The conductive structure **402** comprises a body **402b** and a barrier layer **402a**. The barrier layer **402a** is disposed on sidewalls and a bottom surface of the body **402b**. The barrier layer **402a** of the conductive structure **402** comprises one or more of Ta, TaN, Co, Ru, Al, Ti, TiN, Mn, Nb or air. The body **402b** of the conductive structure **402** comprises one or more of Cu, Ru, W, Ti, Al, Co, Mo, Ir,

(56) Rh.

(57) FIG. 5A, FIG. 5B, FIG. 5C, FIG. 5D, and FIG. 5E illustrate various stages of manufacturing a semiconductor device structure **30**, in accordance with some embodiments of the present disclosure. As shown in FIG. 5A, a substrate **308** is provided. The substrate **308** has a surface **308a** and a surface **308b** opposite the surface **308a**. An isolation layer **309** is formed under the surface **308b**. The isolation layer **309** contacts the surface **308b** of the substrate **308**. The substrate **308** includes electronic devices **310** and **311** embedded in the substrate **308**. The electronic devices **310** and **311** may be of different conductive types. The substrate **308** includes an interconnection structure **312** embedded in the substrate **308**.

(58) Referring to FIG. 5B, a trench **303T** is formed in the substrate **308** and the isolation layer **309** by removing a portion of the substrate **308** and a portion of the isolation layer **309**. A barrier layer **303a** is formed on sidewalls and a bottom surface of the trench **303T**. A material is filled in the trench **303T** to form a body **303b**. The top surfaces of the barrier layer **303a** and the body **303b** are flush with the surface **308a** of the substrate **308**. The barrier layer **303a** and the body **303b** form an interconnection structure **303** in the substrate **308** and the isolation layer **309**. The interconnection structure **303** extends from the surface **308a** of the substrate **308** into the isolation layer **309**. One terminal of the interconnection structure **303** can be exposed by the surface **308a** of the substrate **308**.

(59) Referring to FIG. 5C, an isolation layer **313** is formed on the surface **308a** of the substrate **308**. An isolation layer **314** is formed on the isolation layer **313**. Conductive structures **301**, **302**, and **315** are embedded in the isolation layer **314**. The isolation layers **313** and **314** and the conductive structure **301**, **302**, and **315** form a level L1 of conductive structures on the surface **308a** of the substrate **308**. An isolation layer **316** is formed on a surface **301a** of the conductive structure **301**. The isolation layer **316** is formed on the isolation layer **314**. An isolation layer **317** is formed on the isolation layer **316**. A conductive structure **307** is embedded in the isolation layer **317**. The conductive structure **307** comprises vias **307v1-307v3**. The vias **307v1-307v3** are formed on the surface **301a** of the conductive structure **301**. The vias **307v1-307v3** are in contact with the conductive structure **302**. The vias **307v1-307v3** of the conductive structure **307** covers a first portion of the conductive structure **302** and exposes a second portion of the conductive structure **302**. The isolation layers **316** and **317** and the conductive structure **307** form a level L2 of conductive structures on the surface **308a** of the substrate **308**. The interconnection structure **312** electrically connects the conductive structure **301** and the electronic device **310**.

(60) Referring to FIG. 5D, a portion of the isolation layer **309** on the lower side of the structure shown in FIG. 5C is removed. Also, a portion of the interconnection structure **303** is removed.

(61) Referring to FIG. 5E, an isolation layer **318** is formed under the surface **309a** of the isolation layer **309**. An isolation layer **319** is formed under the isolation layer **318**. Conductive structures

320, 321, and 322 are formed under the surface **309a** of the isolation layer **309**. The conductive structures **320, 321, and 322** are embedded in the isolation layer **319**. The conductive structures **320, 321, and 322** are in contact with the surface **309a** of the isolation layer **309**. The conductive structure **322** contacts the interconnection structure **303**. The isolation layers **318** and **319** and the conductive structures **320, 321, and 322** form a level **L3** of conductive structures under the surface **309a** of the isolation layer **309**.

(62) In the processes of FIGS. 5A-5E, first, the electronic devices **310** and **311** are formed in the substrate **308**. Then the interconnection structure **303** is formed. Afterwards, the conductive structures **301, 302, 315, and 307** are formed above the interconnection structure **303**. Finally, the conductive structures **320, 321, and 322** are formed under the interconnection structure **303**. The interconnection structure **303** is formed before the formation of the conductive structures **301, 302, 315, and 307**. The conductive structures **320, 321, and 322** are formed after the conductive structures **301, 302, 315, and 307**.

(63) FIG. 6A, FIG. 6B, FIG. 6C, and FIG. 6D illustrate various stages of manufacturing a semiconductor device structure **30**, in accordance with some embodiments of the present disclosure. The stages shown in FIGS. 6A-6D indicates a different process for manufacturing a semiconductor device structure **30**.

(64) As shown in FIG. 6A, a substrate **308** is provided. The substrate **308** has a surface **308a** and a surface **308b** opposite the surface **308a**. An isolation layer **309** is formed under the surface **308b**. The isolation layer **309** contacts the surface **308b** of the substrate **308**. The substrate **308** includes electronic devices **310** and **311** embedded in the substrate **308**. The electronic devices **310** and **311** may be of different conductive types. The substrate **308** includes an interconnection structure **312** embedded therein.

(65) An isolation layer **313** is formed on the surface **308a** of the substrate **308**. An isolation layer **314** is formed on the isolation layer **313**. Conductive structures **301, 302** and **315** are embedded in the isolation layer **314**. An isolation layer **316** is formed on the surface **301a** of the conductive structure **301**. The isolation layer **316** is formed on the isolation layer **314**. An isolation layer **317** is formed on the isolation layer **316**. A conductive structure **307** is embedded in the isolation layer **317**. The conductive structure **307** comprises vias **307v1-307v3**. The vias **307v1-307v3** are formed on the surface **301a** of the conductive structure **301**. The vias **307v1-307v3** are in contact with the conductive structure **302**. The vias **307v1-307v3** of the conductive structure **307** covers a first portion of the conductive structure **302** and exposes a second portion of the conductive structure **302**. The interconnection structure **312** electrically connects the conductive structure **301** and the electronic device **310**.

(66) Referring to FIG. 6B, a portion of the isolation layer **309** on the lower side of the structure shown in FIG. 6A is removed.

(67) Referring to FIG. 6C, a trench **303T** is formed in the substrate **308** and the isolation layer **309**. A barrier layer **303a** is formed on sidewalls and a bottom surface of the trench **303T**. A material is filled in the trench **303T** to form a body **303b**. The bottom surfaces of the barrier layer **303a** and the body **303b** are flush with the surface **309a** of the isolation layer **309**. The barrier layer **303a** and the body **303b** form an interconnection structure **303** in the substrate **308** and the isolation layer **309**. The interconnection structure **303** extends from the surface **308a** of the substrate **308** to the surface **309a** of the isolation layer **309**. One terminal of the interconnection structure **303** can be exposed by the surface **309a** of the isolation layer **309**.

(68) Referring to FIG. 6D, an isolation layer **318** is formed under the surface **309a** of the isolation layer **309**. An isolation layer **319** is formed under the isolation layer **318**. Conductive structures **320, 321, and 322** are formed under the surface **309a** of the isolation layer **309**. The conductive structures **320, 321, and 322** are embedded in the isolation layer **319**. The conductive structures **320, 321, and 322** are in contact with the surface **309a** of the isolation layer **309**. The conductive structure **322** contacts the interconnection structure **303**.

(69) In the processes of FIGS. 6A-6D, first, the electronic devices **310** and **311** are formed in the substrate **308**, and the conductive structures **301**, **302**, **315**, and **307** are formed above the substrate **308**. Afterwards, the interconnection structure **303** is formed. Finally, the conductive structures **320**, **321**, and **322** are formed under the interconnection structure **303**. The interconnection structure **303** is formed after the formation of the conductive structures **301**, **302**, **315**, and **307**. The conductive structures **320**, **321**, and **322** are formed after the interconnection structure **303**.

(70) FIG. 7 is a flowchart illustrating a method for manufacturing a semiconductor device, in accordance with various aspects of the present disclosure.

(71) The method **700** includes operation **S701**. In operation **S701**, a substrate including electronic devices embedded therein is provided. The substrate has a first surface and a second surface opposite the first surface. For example, the substrate **308** as shown in FIG. 5A, is provided.

(72) The method **700** includes operation **S702**. In operation **S702**, an isolation layer is formed under the second surface of the substrate. For example, an isolation layer **309** is formed under the surface **308b** of the substrate **308**, as shown in FIG. 5A.

(73) The method **700** includes operation **S703**. In operation **S703**, a first interconnection structure is formed to extend from the first surface of the substrate into the isolation layer. For example, an interconnection structure **303** can be formed to extend from the surface **308a** of the substrate **308** into the isolation layer **309**, as shown in FIG. 5B.

(74) The method **700** includes operation **S704**. In operation **S704**, two levels of conductive structures are formed on the first surface of the substrate. For example, two levels **L1** and **L2** of conductive structures can be formed on the surface **308a** of the substrate **308**, as shown in FIG. 5C.

(75) The method **700** includes operation **S705**. In operation **S705**, a portion of the isolation layer is removed. For example, a portion of the isolation layer **309** can be removed, as shown in FIG. 5D.

(76) The method **700** includes operation **S706**. In operation **S706**, a level of conductive structures is formed under a first surface of the isolation layer. For example, a level **L3** of conductive structures can be formed under a surface **309a** of the isolation layer **309**, as shown in FIG. 5E.

(77) In some embodiments, formation of the two levels of conductive structures comprises forming a second level of conductive structures on a first surface of a first level of conductive structures. For example, the formation of the two levels of conductive structures comprises forming a second level **L2** of conductive structures on a first surface **301a** of a first level **L1** of conductive structures, as shown in FIG. 5C. In some embodiments, formation of an interconnection structure comprises forming a barrier layer on sidewalls and a bottom surface of a via and filling a material in the via. For example, forming an interconnection structure **303** can comprise forming a barrier layer **303a** on sidewalls and a bottom surface of a trench **303T** and filling a material in the trench **303T** to form a body **303b**, as shown in FIG. 5B.

(78) The method **700** is merely an example, and is not intended to limit the present disclosure beyond what is explicitly recited in the claims. Additional operations can be provided before, during, or after each operation of the method **700**, and some operations described can be replaced, eliminated, or reordered for additional embodiments of the method. In some embodiments, the method **700** can include further operations not depicted in FIG. 7.

(79) Some embodiments of the present disclosure provide a semiconductor device. The semiconductor device comprises a substrate, an isolation layer, a first electronic device, a first interconnection structure, a first conductive structure, and a second conductive structure. The substrate has a first surface and a second surface opposite the first surface. The isolation layer contacts the second surface of the substrate and has a first surface facing away from the substrate. The first electronic device is embedded in the substrate. The first interconnection structure extends from the first surface of the substrate to the first surface of the isolation layer. The first conductive structure is disposed on the first surface of the substrate. The second conductive structure is disposed in contact with the first surface of the isolation layer. The first conductive structure and the second conductive structure are electrically connected by the first interconnection structure.

(80) Some embodiments of the present disclosure provide a semiconductor device. The semiconductor device comprises a substrate, an isolation layer, a first interconnection structure, a first conductive structure, and a second conductive structure. The substrate has a first surface and a second surface opposite the first surface. The isolation layer is disposed under the second surface of the substrate and having a first surface. The first interconnection structure has a first end exposed by the first surface of the substrate and a second end exposed by the first surface of the isolation layer. The first conductive structure is disposed on the first surface of the substrate. The second conductive structure is disposed under the first surface of the isolation layer. The first conductive structure and the second conductive structure are electrically connected by the first interconnection structure and configured to receive a first signal.

(81) Some embodiments of the present disclosure provide a method of manufacturing a semiconductor device. The method comprises providing a substrate including electronic devices embedded therein, forming an isolation layer under the second surface of the substrate, forming a first interconnection structure extending from the first surface of the substrate into the isolation layer, forming two levels of conductive structures on the first surface of the substrate, removing a portion of the isolation layer, and forming a level of conductive structures under a first surface of the isolation layer. The substrate has a first surface and a second surface opposite the first surface.

(82) The foregoing outlines structures of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device, comprising: a substrate having a first surface and a second surface opposite the first surface; an isolation layer in contact with the second surface of the substrate and having a first surface facing away from the substrate, wherein the isolation layer is absent above the second surface of the substrate; a first electronic device embedded in the substrate; a first interconnection structure extended from the first surface of the substrate to the first surface of the isolation layer, wherein a width of the first interconnection structure contacting the first surface of the substrate is larger than that contacting the second surface of the substrate; a first conductive structure disposed on the first surface of the substrate; and a second conductive structure disposed in contact with the first surface of the isolation layer, wherein the first conductive structure and the second conductive structure are electrically connected by the first interconnection structure.
2. The semiconductor device of claim 1, further comprising a second interconnection structure embedded in the substrate, wherein the second interconnection structure electrically connects a third conductive structure and the first electronic device.
3. The semiconductor device of claim 1, further comprising a second electronic device embedded in the substrate, wherein the first electronic device and the second electronic device are disposed on different sides of the first interconnection structure.
4. The semiconductor device of claim 2, wherein the third conductive structure and a fourth conductive structure are configured to receive a first reference voltage.
5. The semiconductor device of claim 1, further comprising: a third conductive structure disposed on the first surface of the substrate, wherein the third conductive structure is spaced apart from the first conductive structure.
6. The semiconductor device of claim 5, further comprising: a fourth conductive structure disposed

- under the first surface of the isolation layer, wherein the fourth conductive structure is spaced apart from the second conductive structure.
7. The semiconductor device of claim 1, wherein the first interconnection structure comprises a body and a barrier layer disposed on sidewalls of the body.
8. The semiconductor device of claim 6, further comprising: a fifth conductive structure disposed above a first surface of the first conductive structure, wherein the fifth conductive structure contacts the first conductive structure.
9. The semiconductor device of claim 1, wherein the first conductive structure covers the first interconnection structure from the top view perspective.
10. The semiconductor device of claim 8, wherein the fifth conductive structure includes vias disposed at the intersection of the first conductive structure and the fifth conductive structure.
11. A semiconductor device, comprising: a substrate having a first surface and a second surface opposite the first surface; an isolation layer disposed under the second surface of the substrate and having a first surface, wherein the isolation layer is absent above the second surface of the substrate; a first interconnection structure having a first end exposed by the first surface of the substrate and a second end exposed by the first surface of the isolation layer, wherein a width of the first interconnection structure contacting the first surface of the substrate is larger than that contacting the second surface of the substrate; a first conductive structure disposed on the first surface of the substrate; and a second conductive structure disposed under the first surface of the isolation layer, wherein the first conductive structure and the second conductive structure are electrically connected by the first interconnection structure and configured to receive a first signal.
12. The semiconductor device of claim 11, wherein the first interconnection structure comprises a body and a barrier layer disposed on sidewalls of the body.
13. The semiconductor device of claim 12, wherein the barrier layer comprises one or more of the following materials: Ta, TaN, Co, Ru, Al, Ti, TiN, Mn, Nb or air.
14. The semiconductor device of claim 12, wherein the body comprises one or more of the following materials: Cu, Ru, W, Ti, Al, Co, Mo, Ir, Rh.
15. The semiconductor device of claim 11, wherein a distance from the first surface of the substrate to the first surface of the isolation layer is in a range of about 100 to about 1000 nm.
16. The semiconductor device of claim 11, wherein the first conductive structure comprises a body and a barrier layer disposed on sidewalls and a bottom surface of the body.
17. The semiconductor device of claim 11, further comprising: a third conductive structure disposed on a first surface of the substrate; and a fourth conductive structure disposed under a first surface of the isolation layer, wherein the third conductive structure and the fourth conductive structure are configured to receive a reference voltage.
18. A semiconductor device, comprising: a substrate including electronic devices embedded therein, wherein the substrate having a first surface and a second surface opposite the first surface; an isolation layer under the second surface of the substrate and having a first surface, wherein the isolation layer is absent above the second surface of the substrate; a first interconnection structure extending from the first surface of the substrate into the isolation layer, wherein a width of the first interconnection structure contacting the first surface of the substrate is larger than that contacting the second surface of the substrate; two levels of conductive structures on the first surface of the substrate; and a level of conductive structures under a first surface of the isolation layer.
19. The semiconductor device of claim 18, wherein a second level of the two levels of conductive structures is formed on a first surface of a first level of conductive structures.
20. The semiconductor device of claim 18, wherein the first interconnection structure comprises a body and a barrier layer disposed on sidewalls of the body.
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